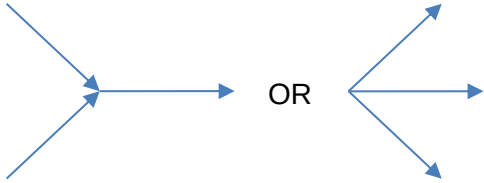


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		Vector diagram roughly as shown (1)  Vertical components cancel (1) Accept forces due to +ve charges cancel Field is right OR attractive force towards positives (1) Electron is negative linked to F (or a) left OR repulsive from -ve (1)		4		4	2	
	(b)		Use of $F = \frac{kQq}{r^2}$ OR $F = EQ$ and $E = \frac{kq}{r^2}$ (1) (expect to see 9.6×10^{-18} [N]) $2E\cos 60 + E$ or $2E\sin 30 + E$ OR $2F\cos 60 + F$ or $2F\sin 30 + F$ (1) (expect to see 1.92×10^{-17} [N] or 120 [N C⁻¹]) Use of $F = ma$ (1) (can be awarded for any valid force $\div 9.11 \times 10^{-31}$) Correct answer = 2.1×10^{13} [m s⁻²] (no ecf for this, must be correct) (1)	1	1		4	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Substitution into $PE = \frac{kQq}{r}$ e.g. $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times 1.6 \times 10^{-19}}{0.6}$ (1) $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6} + \frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6} +$ $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times 1.6 \times 10^{-19}}{0.6}$ seen or equivalent or $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6}$ seen with statement that the other 2 potentials cancel (1)	1					
					1		2	1	
	(d)		Valid method e.g. considering potential energy/potential (1) New PE = $2 \times -5.76 \times 10^{-18} + 2.88 \times 10^{-18} = -8.64 \times 10^{-18}$ [J] OR new potential = 54 [V] OR PE must be lower because -ve values are the same but the positive is smaller (doesn't need a calculation) OR new potential is higher than original because +ve potentials are the same but negative is smaller (36V / 37.5 [V] to 54 [V]) (1) Can travel to lower PE OR will gain KE OR can travel to higher potential because electron is negative (1) Correct conclusion based on correct physics (yes is sufficient if the explanation is clear) (1) Alternative based on forces: Attractive forces (or forces due to +ve charges) are the same/symmetrical for both points OR oscillating with SHM if attractive forces only OR would just reach R with attractive forces only OR force at R is 7.2×10^{-18} [N] ($\rightarrow$) (1) Repulsive force / force due to negative charge helps OR repulsive force at end is 2.4×10^{-18} [N] OR force at P is 19.2×10^{-18} [N] ($\leftarrow$) (1) Overall of these is to the left (1) Conclusion – yes (1)			4	4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Award 3 marks for: Yes it gets there with 16.5 / 18 eV of KE (with no reasoning) Yes it gets there with $2.88 / 2.64 \times 10^{-18}$ J of KE (with no reasoning) Yes it gets there with a speed of 2.5×10^6 m s⁻¹ (with no reasoning) Award 2 marks for: Yes it gets there because the force due to -2.4 nC always repels Yes because acceleration left is greater than acceleration right Award 1 mark for: Yes / don't know / No because positive charges provide force to right at the end.						
			Question 3 total	3	7	4	14	8	0